

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

IV Semester of BSc (Biological Sciences - BT/MI) Examination May 2018

Course code & Name: MI 402 General Virology

Date: 07/05/2018 Day: Monday Time: 10:00 A M To 10:20 A M Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose the correct answer for the following questions.

10

1. Bacteriophage having head and tail is an example of the following virus
(i) polyhedral (ii) enveloped (iii) complex (iv) helical
2. Viruses multiply only within the living host cells because they are the following
(i) obligate intracellular parasites (ii) facultative pathogen
(iii) anaerobic (iv) pathogens
3. The following indicates viral infection that lead to disease in the host
(i) lysogenic cycle (ii) lytic cycle (iii) acute infection (iv) all of the above
4. The information for the biosynthesis of diphtheria toxin in *Corynebacterium diphtheriae* is encoded on the following
(i) bacterial genome (ii) plasmid (iii) prophage (iv) viral RNA
5. Most Quasi-species viruses arise from the following viruses
(i) complex (ii) DNA (iii) enveloped (iv) RNA

6. The following viruses lack the full complement of genes necessary for a complete infectious cycle
(i) temperate (ii) ssRNA (iii) dsRNA (iv) defective
7. The following usually do not infect plants
(i) viruses (ii) prions (iii) viroids (iv) bacteria
8. The following RNA viruses form ds DNA intermediates during their reproduction
(i) *Retroviridae* (ii) *Hepadnaviridae* (iii) *Herpesviridae* (iv) *Reoviridae*
9. The following viruses contain both (+) and (-) strand information on a single strand of RNA.
(i) nonsense (ii) missense (iii) ambisense (iv) none of the above
10. The following giant virus was the first to be discovered among all giant viruses
(i) vaccinia (ii) pandora (iii) mimi (iv) bacteriophage

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IV Semester of BSc (Biological Sciences - BT/MI) Examination May 2018

Course code & Name: MI 402 General Virology

Date: 07/05/2018 Day: Monday Time: 10:20 A M To 12:00 P M Maximum Marks: 25

Instructions:

1. Section I and II must be attempted in SINGLE ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION - I

- Q - II Answer the following questions as directed 10**
- A Define any two the following terms (02)**
- i. Satellite viruses
 - ii. Quasi-species viruses
 - iii. Persistent viral infection
- B Discuss any three of the following: (06)**
- i. Evolution of Measles viruses
 - ii. Recombination of viruses by break-rejoin mechanism
 - iii. Giant viruses
 - iv. Production of vaccines against viruses
- C Discuss briefly Replication of viroid with diagram (02)**

OR

Discuss briefly Replication of Gapped DNA viruses with diagram

SECTION - II

- Q-III Answer the following questions as directed 15**
- A Discuss briefly any two of the following (04)**
- i. General Features of Virus Replication
 - ii. Classification of viruses on the basis of its morphology
 - iii. One-step growth curve of viruses with neatly labeled diagram
- B Differentiate between any three of the following: (03)**
- i. Bacteria and viruses
 - ii. Bacteriophage and temperate phage
 - iii. Baltimore classification and taxonomic classification of viruses
 - iv. Viroid and prion
- C Discuss briefly any one of the following (03)**
- i. The lytic cycle of a bacteriophage
 - ii. Entry of animal virus into its host cell
- D Discuss briefly any two of the following (04)**
- i. Specialized transduction
 - ii. Coding Strategies in viruses
 - iii. Viral Shunt
- E Justify: Treatment of viral diseases using antibiotics is difficult (01)**